

Exploit Development Basics

Introduction

Exploit development is the process of creating techniques or code that use vulnerabilities to trigger unintended behavior such as crashing software, bypassing authentication, or executing commands.

Security researchers study exploit development to understand attacker methods and improve defenses.

Common Vulnerability Types

Buffer Overflow

Occurs when software writes more data than memory buffer size.

Possible impact:

- Application crash
- Code execution
- System compromise

Common in older C/C++ applications.

SQL Injection

Occurs when unsanitized input is inserted into database queries.

Example payload:

```
' OR '1'='1
```

Impact:

- Login bypass
- Read data
- Modify records
- Delete tables

Cross-Site Scripting (XSS)

Attacker injects JavaScript executed in victim browser.

Example:

```
<script>alert('XSS')</script>
```

Impact:

- Session theft
- Phishing
- Defacement

Command Injection

Application passes unsafe input to system commands.

Impact:

- Remote command execution

Path Traversal

Using input like ../ to access unintended files.

Impact:

- Read configuration files
- Read credentials

Basic Exploit Development Workflow

1. Identify vulnerability
2. Reproduce crash or flaw
3. Understand root cause
4. Create proof of concept

5. Test safely in lab
6. Document impact

Tools Commonly Used

- Python
- Exploit Database
- GDB
- Immunity Debugger

Security Mitigations

ASLR

Randomizes memory addresses.

Makes exploitation harder.

DEP / NX

Prevents execution from non-executable memory regions.

WAF

Filters malicious web requests.

Patching

Most effective long-term solution.

Real-Life Case Study – SQL Injection

An e-commerce site accepted raw input in login form.

Payload:

admin' --

Result:

Authentication bypass.

Fix:

Prepared statements and input validation.

Real-Life Case Study – XSS in Comment Section

User comments were rendered without sanitization.

Impact:

Attacker stole session cookies from visitors.

Fix:

Output encoding, CSP headers, sanitization.

Conclusion

Vulnerability scanning helps discover weaknesses, penetration testing proves real attack paths, and exploit development explains how vulnerabilities are abused. Together, these three areas form a strong foundation for modern cybersecurity assessments and help organizations reduce real-world risk.